A complete log-concavity classification for higher-order Stirling cycle rows

Carptopus
carptopus@163.com

Manuscript, 2 September 2026

Abstract

Let $[N, k]_{\geq r}$ denote the number of permutations of N elements having exactly k cycles, every cycle of length at least r, and define the higher-order Stirling cycle rows

$$C_r(n, k) = [n + (r-1)k, k]_{\geq r}, \qquad 0 \leq k \leq n.$$

Deb and Sokal proved that these rows are log-concave for $r = 2$ and asked whether the same holds for $r = 3, 4, 5$. We prove all three remaining cases. Together with the elementary $r = 1$ case, this gives a complete classification: every row $C_r(n, \cdot)$ is log-concave for every n if and only if $1 \leq r \leq 5$. The cutoff is already visible in the row $n = 3$; an explicit ratio is strictly decreasing in r, is greater than one at $r = 5$, and is less than one at $r = 6$.

The proofs use three different mechanisms. The case $r = 3$ follows from Sagan's inductive criterion. For $r = 4$ we introduce a factorial normalization and prove that the recurrence preserves the narrower weighted log-concavity cone determined by the removed factorials. For $r = 5$ we combine two exact parameter wedges, a certified finite prefix of $5, 754, 989$ inequalities, and an effective fifth-order Fourier–Edgeworth estimate on the remaining compact band. We also prove a uniform structural result valid for every $r \geq 2$: the normalized rows

$$\frac{k!\, C_r(n, k)}{(n + (r-1)k)!}$$

are log-concave.

1 Introduction

For integers $r \geq 1$, let $[N, k]_{\geq r}$ be the signless r-associated Stirling number of the first kind. Thus $[N, k]_{\geq r}$ counts permutations of an N-element set having k cycles, with every cycle of length at least r. The higher-order Stirling cycle triangle studied by Deb and Sokal [1] has entries

$$C_r(n, k) = [n + (r-1)k, k]_{\geq r}. \tag{1.1}$$

The unusual feature of (1.1) is that the size of the ground set changes with k along a fixed row. Consequently, real-rootedness results for the polynomial $\sum_k [N, k]_{\geq r} x^k$ with fixed N do not imply log-concavity of the rows in (1.1). Indeed, Deb and Sokal showed that the row-generating polynomials in (1.1) generally have nonreal zeros when $r \geq 3$ [1].

Deb and Sokal proved row log-concavity for $r = 2$ and conjectured it for $r = 3, 4, 5$ [1, Conjecture 1.3(c)]. The same three cases were recorded as Problem 30.9 in the BCC30 problem collection [2]. Our main result settles those remaining parameter layers and also locates the exact cutoff.

Theorem 1.1 (complete classification). For an integer $r \geq 1$, the sequence

$$C_r(n,0), C_r(n,1), \ldots, C_r(n,n)$$

is log-concave for every $n \geq 0$ if and only if $1 \leq r \leq 5$.

Here and below, entries outside $0 \leq k \leq n$ are zero. The positive part of each row is contiguous, so the usual adjacent inequalities are sufficient.

The proof naturally separates into four pieces. The ordinary case $r = 1$ follows from

$$\sum_k [n,k]_{\geq 1} x^k = x(x+1)\cdots(x+n-1),$$

and $r = 2$ is due to Deb and Sokal. Section 3 proves $r = 3$ by applying Sagan's log-concavity-preserving recurrence criterion [3]. Section 4 proves $r = 4$ by preserving a weighted cone rather than the full cone of log-concave sequences. Sections 6–10 give the $r = 5$ proof. Section 5 shows that every $r \geq 6$ fails already for $n = 3$.

One structural result applies at every order and explains why factorial normalization is useful even when the original rows cease to be log-concave.

Theorem 1.2 (normalized rows). Let $r \geq 2$ and put

$$H_r(n,k) = \frac{k!\, C_r(n,k)}{(n+(r-1)k)!}. \tag{1.2}$$

For every $n \geq 0$, the sequence $H_r(n,0), \ldots, H_r(n,n)$ is log-concave.

The $r = 5$ part of Theorem 1.1 is computer-assisted. Its finite and symbolic components use exact rational arithmetic or decimal arithmetic with a proved relative-rounding bound. The analytic tail is reduced to explicit rational inequalities. Section 11 records the verification boundary and the reproducibility entry points.

2 Recurrences and an all-order normalization

Write $p = r - 1$ and

$$N = n + pk.$$

The standard insertion recurrence for associated Stirling numbers becomes

$$C_r(n,k) = (N-1)C_r(n-1,k) + (N-1)_{r-1}C_r(n-1,k-1), \tag{2.1}$$

where $(x)_j = x(x-1)\cdots(x-j+1)$. The first term inserts the new element into an existing cycle. The second term forms a new r-cycle from the new element and $r-1$ additional elements.

After the normalization (1.2), all falling factorials cancel and (2.1) becomes

$$H_r(n,k) = a_k H_r(n-1,k) + b_k H_r(n-1,k-1), \tag{2.2}$$

with

$$a_k = \frac{n + pk - 1}{n + pk}, \qquad b_k = \frac{k}{n + pk}. \tag{2.3}$$

We use the following form of Sagan's criterion [3, Theorem 1]. Suppose a nonnegative triangular array satisfies

$$y_k = d_k x_k + c_k x_{k-1}.$$

If c_k and d_k are log-concave in k and

$$2c_k d_k \geq c_{k-1} d_{k+1} + c_{k+1} d_{k-1}, \tag{2.4}$$

then the transformation preserves log-concavity. Although Sagan states the theorem for nonnegative integer arrays, its proof uses only addition, multiplication, and nonnegative inequalities, so it applies without change to nonnegative rational arrays.

Proof of Theorem 1.2. For the coefficients in (2.3), direct simplification gives, for $n \geq 2$ and $1 \leq k \leq n - 1$,

$$a_k^2 - a_{k-1}a_{k+1} = \frac{p^2(2kp + 2n - 1)}{(kp + n)^2(kp + n - p)(kp + n + p)}, \tag{2.5}$$

$$b_k^2 - b_{k-1}b_{k+1} = \frac{n(2kp + n)}{(kp + n)^2(kp + n - p)(kp + n + p)}, \tag{2.6}$$

and

$$2a_k b_k - a_{k-1}b_{k+1} - a_{k+1}b_{k-1} = \frac{2p(knp + kp + n^2)}{(kp + n)^2(kp + n - p)(kp + n + p)}. \tag{2.7}$$

All three quantities are nonnegative in the stated range. The zero boundaries are immediate, and the initial row consists only of $H_r(0, 0) = 1$. Sagan's criterion applied successively to (2.2) proves the theorem. □

The removed factor

$$W_r(n, k) = \frac{(n + pk)!}{k!} \tag{2.8}$$

is generally log-convex in k. Theorem 1.2 therefore does not imply log-concavity of $C_r(n, \cdot)$, but it provides the common base sequence used in the $r = 4$ argument.

3 The third-order rows

Theorem 3.1. For every $n \geq 0$, the row $C_3(n, \cdot)$ is log-concave.

Proof. In (2.1), put $r = 3$ and

$$L = n + 2k - 1.$$

Then

$$C_3(n,k) = LC_3(n-1,k) + L(L-1)C_3(n-1,k-1). \tag{3.1}$$

In Sagan's notation, take

$$c_k = L(L-1), \qquad d_k = L.$$

The sequence d_k is a nonnegative arithmetic progression. The sequence c_k is the pointwise product of two nonnegative arithmetic progressions. Both are log-concave on the required index range. The mixed defect in (2.4) is

$$2d_kc_k - d_{k-1}c_{k+1} - d_{k+1}c_{k-1} = 8(L-1) \geq 0. \tag{3.2}$$

The initial row and the zero boundaries satisfy the claim, so induction using Sagan's criterion completes the proof. □

4 The fourth-order rows

The direct use of Sagan's criterion fails at $r = 4$: its mixed coefficient defect has the wrong sign. We retain the factorial normalization but impose the stronger inequality needed after restoring the weights.

For this section set $p = 3$, $N = n + 3k$, and write

$$W(n,k) = \frac{N!}{k!}, \qquad H(n,k) = \frac{C_4(n,k)}{W(n,k)}.$$

The original row is log-concave exactly when

$$H(n,k)^2 \geq T(n,k)H(n,k-1)H(n,k+1), \tag{4.1}$$

where

$$T(n,k) = \frac{W(n,k-1)W(n,k+1)}{W(n,k)^2} = \frac{k}{k+1}\prod_{i=1}^{3}\frac{N+i}{N-3+i}. \tag{4.2}$$

Theorem 4.1. For every $n \geq 0$, the row $C_4(n,\cdot)$ is log-concave.

Proof. Assume the preceding row $x_j = H(n-1,j)$ satisfies (4.1). First work on four consecutive entries in its positive support and set

$$\rho_j = \frac{x_j}{x_{j-1}}.$$

At positive entries, two adjacent weighted inequalities give

$$\rho_{k-1} \geq T(n-1,k-1)\rho_k, \qquad \rho_{k+1} \leq \frac{\rho_k}{T(n-1,k)}. \tag{4.3}$$

The recurrence (2.2) has

$$a_j = \frac{n+3j-1}{n+3j}, \qquad b_j = \frac{j}{n+3j}.$$

After dividing the desired defect in row n by x_{k-1}^2, its right-hand side is increasing in $1/\rho_{k-1}$ and in ρ_{k+1}. Thus the two equality cases in (4.3) give the worst possible value. With $\rho = \rho_k$, it is enough to prove

$$(a_k\rho + b_k)^2 \geq T(n,k)\rho \left(a_{k-1} + \frac{b_{k-1}}{T(n-1,k-1)\rho}\right)\left(\frac{a_{k+1}\rho}{T(n-1,k)} + b_{k+1}\right). \tag{4.4}$$

Put $M = n + 3k$. Clearing the positive denominator

$$M^2(M-3)^2(M-2)^2(M-1)^2$$

turns the left side minus the right side of (4.4) into a quadratic polynomial $P(\rho)$. Its leading coefficient is

$$9(M-3)^2(M-2)^2(M-1)^2 > 0,$$

and its discriminant is

$$-972k^2(M-4)^2(M-3)^2(M-2)^3(M-1)^2(M+2) < 0. \tag{4.5}$$

Hence $P(\rho) > 0$ for every $\rho > 0$. This proves the internal inequalities for $n \geq 3$ and $2 \leq k \leq n-1$. The case $k = 1$ is immediate because $C_4(n,0) = 0$. At the left endpoint, the same expanded inequality is obtained as the limit $x_0 \downarrow 0$, for which $1/\rho_1 \downarrow 0$; this remains below the upper bound in (4.3). At the right endpoint, the limit $x_n \downarrow 0$ gives $\rho_n \downarrow 0$, again no larger than the worst-case substitution. The short initial rows are direct. Induction proves the theorem. $\square$

The negative discriminant in (4.5) is specific to the weighted cone at $p = 3$. At $p = 4$ the analogous discriminant is positive, and actual rows enter the interval where the quadratic is negative. The fifth-order case therefore needs a different argument.

5 The sharp cutoff after order five

The first nontrivial three-term row already determines the failure side of the classification.

Proposition 5.1. For $r \geq 2$, the row $C_r(3,\cdot)$ is log-concave if and only if $2 \leq r \leq 5$. In particular, the assertion that every row is log-concave fails for every $r \geq 6$.

Proof. The three positive entries are

$$C_r(3,1) = (r+1)!,$$

$$C_r(3,2) = \frac{(2r+1)!}{r(r+1)},$$

and

$$C_r(3,3) = \frac{(3r)!}{6r^3}.$$

Thus the only nontrivial log-concavity ratio is

$$R(r) = \frac{C_r(3,2)^2}{C_r(3,1)C_r(3,3)} = \frac{6r(2r+1)!^2}{(r+1)^2(3r)!(r+1)!}. \tag{5.1}$$

Its adjacent ratio is

$$\frac{R(r+1)}{R(r)} = \frac{4(r+1)^4(2r+3)^2}{3r(r+2)^3(3r+1)(3r+2)}. \tag{5.2}$$

The denominator of (5.2) minus its numerator equals

$$11r^6 + 77r^5 + 168r^4 + 80r^3 - 136r^2 - 144r - 36.$$

After substituting $r = s + 1$, this becomes

$$11s^6 + 143s^5 + 718s^4 + 1742s^3 + 2047s^2 + 947s + 20,$$

which is positive for $s \geq 0$. Hence $R(r)$ is strictly decreasing for $r \geq 1$. Finally,

$$R(5) = \frac{55}{39} > 1, \qquad R(6) = \frac{5148}{5831} < 1.$$

This proves the proposition. □

Deb and Sokal's positive conjecture stops at $r = 5$. Proposition 5.1 does not refute a claim they made for $r \geq 6$; it shows that their stated endpoint is the sharp endpoint of the all-row phenomenon.

6 Reduction of the fifth-order case

It remains to prove the last positive layer.

Theorem 6.1. For every $n \geq 0$, the row $C_5(n, \cdot)$ is log-concave. At every internal index for which the three terms are positive, the inequality is strict.

Write

$$d = n - k, \qquad f(z) = \sum_{j \geq 0} \frac{z^j}{j+5}, \qquad B(k,d) = [z^d]f(z)^k. \tag{6.1}$$

The exponential formula for permutations by cycles gives

$$C_5(n,k) = \frac{(5k+d)!}{k!} B(k,d). \tag{6.2}$$

For $k \geq 2$, put

$$\mu = \frac{d}{k}, \qquad q = \frac{1}{k}.$$

After substituting (6.2) at $k-1, k, k+1$, the required inequality is equivalent to

$$B(k,d)^2 \geq T_k(\mu)B(k-1,d+1)B(k+1,d-1), \tag{6.3}$$

where

$$T_k(\mu) = \frac{1}{1+q}\prod_{i=1}^{4}\frac{\mu+5+iq}{\mu+5-(i-1)q}. \tag{6.4}$$

The proof of (6.3) divides the parameter plane into two exact wedges and one compact band.

7 Two exact wedges

Define

$$Q_d(k) = d!\,5^k B(k,d). \tag{7.1}$$

The identity

$$(1-z)\big(zf'(z)+5f(z)\big) = 1$$

implies

$$Q_d(k) = \frac{d(5k+d-1)}{5k+d}Q_{d-1}(k) + \frac{5k}{5k+d}Q_d(k-1). \tag{7.2}$$

To state the two-step lemma with the correct row index, put

$$q_n(k) = Q_{n-k}(k). \tag{7.3}$$

Then (7.2) is the triangular recurrence

$$q_n(k) = A(n,k)q_{n-1}(k) + D(n,k)q_{n-1}(k-1), \tag{7.4}$$

where

$$A(n,k) = \frac{(n-k)(n+4k-1)}{n+4k}, \qquad D(n,k) = \frac{5k}{n+4k}. \tag{7.5}$$

Lemma 7.1 (two-step preservation). Two successive applications of (7.4), from the row indexed by $n-2$ to the row indexed by n, preserve log-concavity at every internal index with $k \geq 2$ and $d = n-k \geq 3$. The boundary defects corresponding to $d = 1, 2$ are strictly positive. Starting from the short initial rows, every row $q_n(\cdot)$ is therefore log-concave.

Proof certificate. Write the two-step map as

$$y_k = \alpha_k x_k + \beta_k x_{k-1} + \gamma_k x_{k-2}.$$

For a log-concave input, write four consecutive ratios as $u \geq v \geq w \geq t$. Direct differentiation of the output defect shows that its minimum occurs at $u = v$ and $t = w$. Put $v = w + h$ with

$h \geq 0$. After substituting $k = K + 2$ and $d = D + 3$, the numerator of the defect has exactly twelve monomials in (h, w). Eleven coefficients are coefficientwise nonnegative in (K, D). In the h-free part, the only coefficient not individually nonnegative is the coefficient c_1 of w; the exact identity

$$4c_0c_2 - c_1^2 \geq 0$$

is coefficientwise nonnegative in (K, D). Since the coefficient c_2 and the common denominator are positive, the quadratic $c_0 + c_1 w + c_2 w^2$ is nonnegative for $w > 0$. This establishes the internal part of the lemma over exact rational functions. For $d = 1, 2$, the defects are respectively

$$\frac{5(33k + 2)}{252}$$

and

$$\frac{25(1155k^3 + 2426k^2 + 1435k + 168)}{63504}.$$

The full twelve-coefficient expansion and the shifted polynomial $4c_0c_2 - c_1^2$ are regenerated exactly by the verification program listed in Section 11. This is a finite symbolic certificate for the displayed algebraic reduction, not a sample of (n, k) values.

We next connect Lemma 7.1 to (6.3). Since $d = n - k$,

$$q_n(k)^2 \geq q_n(k-1)q_n(k+1)$$

is exactly

$$Q_d(k)^2 \geq Q_{d+1}(k-1)Q_{d-1}(k+1). \tag{7.6}$$

Using (7.1), inequality (6.3) follows from (7.6) whenever

$$W(k, d) = \frac{d}{d+1} T_k(d/k) \leq 1. \tag{7.7}$$

Explicitly,

$$W(k, d) = \frac{dk(d + 5k + 1)(d + 5k + 2)(d + 5k + 3)(d + 5k + 4)}{(d + 1)(k + 1)(d + 5k)(d + 5k - 1)(d + 5k - 2)(d + 5k - 3)}. \tag{7.8}$$

Let $P(k, d)$ be the numerator of $1 - W(k, d)$ after clearing the positive denominator. The three substitutions needed below give polynomials with nonnegative coefficients:

$$\begin{aligned} P(2(U+1) + 1 + S, U + 1) = {} & 625S^5 + 5375S^4U + 8375S^4 \\ & + 17450S^3U^2 + 54100S^3U + 41925S^3 \\ & + 25630S^2U^3 + 117810S^2U^2 + 180885S^2U + 93025S^2 \\ & + 15125SU^4 + 89540SU^3 + 200255SU^2 + 201630SU + 77450S \\ & + 1331U^5 + 6655U^4 + 11495U^3 + 7865U^2 + 2054U + 600, \end{aligned} \tag{7.9}$$

$$P(2(U+12),U+12) = 1331U^5 + 64735U^4 + 1185415U^3 + 9744665U^2 + 31222854U + 8690040, \tag{7.10}$$

and, for $k = K+3$, $d = 12(K+3)+S$,

$$\begin{aligned} P(k,d) = {} & 142477K^5 + 93925K^4S + 1754230K^4 + 18530K^3S^2 \\ & + 1032920K^3S + 8228255K^3 + 1610K^2S^3 + 158220K^2S^2 \\ & + 4225545K^2S + 17796230K^2 + 65KS^4 + 9320KS^3 \\ & + 449165KS^2 + 7608500KS + 16368048K \\ & + S^5 + 190S^4 + 13475S^3 + 423830S^2 + 5076504S + 3630240. \end{aligned} \tag{7.11}$$

Thus $W < 1$ for $k \geq 2d+1$, for the boundary $k = 2d$ when $d \geq 12$, and for $d \geq 12k$ when $k \geq 3$. The remaining points on $k = 2d$ have $d \leq 11$ and are covered by exact all-k strip certificates through $d = 18$.

The low-density wedge also needs the internal index $k = 2$, which is not supplied by (7.11). We record its direct proof.

Lemma 7.2 (the $k = 2$ line). The fifth-order row inequality at $k = 2$ holds for every $n \geq 3$.

Proof. Put

$$A = \sum_{j=5}^{n+3} \frac{1}{j}, \qquad e = \sum_{j=n+4}^{n+7} \frac{1}{j}.$$

Cycle decomposition gives

$$C_5(n,1) = (n+3)!, \qquad C_5(n,2) = (n+7)!\,A,$$

and

$$C_5(n,3) = (n+11)!\,S_3,$$

where

$$S_3 = \sum_{\substack{5 \leq i < j \leq n+7 \\ j-i \geq 5}} \frac{1}{ij}.$$

Dropping the separation condition gives

$$S_3 < \frac{(A+e)^2}{2}.$$

Consequently, the desired inequality follows from

$$2A^2 \geq R(A+e)^2, \qquad R = \prod_{j=0}^{3} \frac{n+8+j}{n+4+j}. \tag{7.12}$$

Set $x = n+4$. For $x \geq 29$,

$$A \geq \log(x/5), \qquad e \leq \frac{4}{x}, \qquad \sqrt{R} \leq \left(1+\frac{4}{x}\right)^2.$$

The two factors in

$$\left(1+\frac{4}{x}\right)^2 \left(1+\frac{4}{x\log(x/5)}\right)$$

decrease with x. At $x = 29$, use $\log(29/5) > 7/4$ and $\sqrt{2} > 7/5$ to obtain

$$\left(\frac{33}{29}\right)^2 \left(1+\frac{16}{203}\right) = \frac{238491}{170723} < \frac{7}{5} < \sqrt{2}.$$

This is (7.12) for every $n \geq 25$. The logarithmic bound follows from a rational Taylor upper bound for $\exp(7/4)$; the cases $3 \leq n \leq 24$ are checked exactly over the rationals. □

Lemma 7.3 (exact wedges). Inequality (6.3) holds strictly in each of the following regions:

$$k \geq 2d, \tag{7.13}$$

and

$$d \geq 12k. \tag{7.14}$$

For the first wedge, (7.9) handles $k \geq 2d+1$, (7.10) handles $k = 2d$ for $d \geq 12$, and the strip certificates handle the remaining boundary points. For the second wedge, (7.11) handles $k \geq 3$, $k = 1$ is a zero-boundary inequality, and Lemma 7.2 handles $k = 2$. In each symbolic region $W < 1$, so positivity of the neighboring Q-entries makes the original inequality strict. All boundary exceptions are therefore included rather than inferred from an asymptotic estimate.

The only region left after Lemma 7.3 is

$$\frac{1}{2} < \mu < 12. \tag{7.15}$$

8 A common saddle point and the compact-band margin

For each μ in (7.5), there is a unique $z \in (0,1)$ satisfying

$$\mu = \frac{zf'(z)}{f(z)}. \tag{8.1}$$

Define a span-one probability distribution by

$$\Pr_z(J = j) = \frac{z^j}{(j+5)f(z)}, \qquad j \geq 0. \tag{8.2}$$

Let σ^2 and κ_s be its variance and cumulants. If S_m is a sum of m independent copies of J, then

$$B(k,d) = f(z)^k z^{-d} \Pr_z(S_k = d). \tag{8.3}$$

The same z works for all three coefficients in (6.3). The central term $S_k = d$ is at its mean. The other two probabilities have displacements

$$S_{k-1} = d + 1 : \quad \mu + 1, \qquad S_{k+1} = d - 1 : \quad -(\mu + 1) \tag{8.4}$$

from their respective means. All exponential saddle factors in (8.3) therefore cancel.

Put

$$\Delta(k, \mu) = \log \frac{B(k, d)^2}{T_k(\mu) B(k-1, d+1) B(k+1, d-1)}. \tag{8.5}$$

The first-order coefficient in the common-saddle expansion is

$$M(\mu) = \frac{(\mu+1)^2}{\sigma^2} + 1 - \frac{16}{\mu+5}. \tag{8.6}$$

It is positive precisely because the saddle variance satisfies

$$\sigma^2(11 - \mu) < (\mu+1)^2(\mu+5). \tag{8.7}$$

The verification of (8.7) reduces, after writing f as its elementary logarithmic expression, to a rational inequality whose differentiated remainder is a nonnegative square. On the full compact band, a rational Bernstein certificate sharpens it to

$$M(\mu) > \frac{3}{100}. \tag{8.8}$$

Expanding the three local probabilities through order five gives

$$\Delta(k, \mu) = \frac{M}{k} + \frac{C_2}{k^2} + \frac{C_3}{k^3} + \frac{C_4}{k^4} + \frac{C_5}{k^5} + R_k. \tag{8.9}$$

Each C_j is a rational expression in z, $f(z)$, and the cumulants through order ten. Exact two-variable Bernstein certificates on the complete saddle band prove

$$C_2 > -2M, \qquad C_3 > 0, \qquad C_4 > -4M, \qquad C_5 > 0. \tag{8.10}$$

Hence the retained part of (8.9) is at least

$$\frac{3}{100}\left(\frac{1}{k} - \frac{2}{k^2} - \frac{4}{k^4}\right). \tag{8.11}$$

9 Effective Fourier control for the infinite tail

We now give the effective local expansion used to bound R_k. Let

$$\phi_z(t) = \frac{f(ze^{it})}{f(z)}, \qquad \psi_z(t) = e^{-i\mu t}\phi_z(t),$$

and normalize the lattice probabilities by

$$\overline{p}_m(b) = \sigma\sqrt{2\pi m}\,\Pr_z(S_m = m\mu + b). \tag{9.1}$$

Fourier inversion with

$$u = \sigma\sqrt{m}\,t, \qquad \varepsilon = m^{-1/2}$$

gives

$$\overline{p}_m(b) = \frac{1}{\sqrt{2\pi}}\int_{-\pi\sigma\sqrt{m}}^{\pi\sigma\sqrt{m}} e^{-ibu/(\sigma\sqrt{m})}\psi_z\left(\frac{u}{\sigma\sqrt{m}}\right)^m du. \tag{9.2}$$

The Gaussian term must be extracted before estimating the correction. On the local analytic disk,

$$e^{-ibu/(\sigma\sqrt{m})}\psi_z\left(\frac{u}{\sigma\sqrt{m}}\right)^m = e^{-u^2/2}\exp\left(\sum_{j\geq 1} E_j(u,b)\varepsilon^j\right), \tag{9.3}$$

where

$$E_1(u,b) = -\frac{ibu}{\sigma} + \frac{\kappa_3(iu)^3}{6\sigma^3},$$

and, for $2 \leq j \leq 8$,

$$E_j(u,b) = \frac{\kappa_{j+2}(iu)^{j+2}}{(j+2)!\sigma^{j+2}}. \tag{9.4}$$

The three displacements needed in (8.5) satisfy $|b| \leq \mu + 1$. Exact rational certificates over the saddle band prove

$$\frac{\mu+1}{\sigma} < \frac{7}{4}$$

and the termwise bounds

$$|E_j(u,b)| \leq A_j(u), \tag{9.5}$$

where

$$\begin{aligned}
A_1 &= \frac{7}{4}|u| + \frac{11}{20}|u|^3, & A_2 &= \frac{13}{20}|u|^4,\\
A_3 &= \frac{17}{20}|u|^5, & A_4 &= \frac{6}{5}|u|^6,\\
A_5 &= \frac{7}{4}|u|^7, & A_6 &= \frac{5}{2}|u|^8,\\
A_7 &= \frac{15}{4}|u|^9, & A_8 &= \frac{11}{2}|u|^{10}.
\end{aligned} \tag{9.6}$$

Define the exact local polynomials $H_s(u,b)$ by

$$\exp\left(\sum_{j=1}^{8} E_j(u,b)\varepsilon^j\right) = \sum_{s\geq 0} H_s(u,b)\varepsilon^s.$$

Equating coefficients gives the recurrence

$$sH_s = \sum_{j=1}^{\min(8,s)} jE_jH_{s-j}. \tag{9.7}$$

The five probability coefficients used in Section 8 are therefore

$$p_j(b) = \frac{1}{\sqrt{2\pi}}\int_{\mathbb{R}} e^{-u^2/2}H_{2j}(u,b)\,du, \qquad 1 \leq j \leq 5. \tag{9.8}$$

The odd integrated terms vanish. Substituting $b = 0, \mu+1, -(\mu+1)$ at sample sizes $k, k-1, k+1$, expanding $(k \pm 1)^{-j}$ in $1/k$, and including the normalizing and target-weight terms produces exactly the coefficients $M, C_2, \ldots, C_5$ in (8.9).

To control cumulants above order ten, write

$$5f(z) = \frac{1}{1-C(z)}, \qquad C(z) = \sum_{\ell\geq 1} c_\ell z^\ell,$$

where $c_\ell > 0$ and $\sum_{\ell\geq 1} c_\ell = 1$. Define

$$\log(5f(z)) = \sum_{\ell\geq 1} \beta_\ell z^\ell,$$

For completeness, both assertions about C follow directly from the defining function. Differentiation gives

$$z(1-z)C'(z) = 5(1-C(z))(z-C(z)).$$

Writing $C(z) = \sum_{m\geq 1} c_m z^m$ yields

$$(m+5)c_m = (m-6)c_{m-1} + 5\sum_{i=1}^{m-1} c_i c_{m-i}, \qquad m \geq 2. \tag{9.9}$$

The first five coefficients are

$$\frac{5}{6}, \quad \frac{5}{252}, \quad \frac{5}{378}, \quad \frac{605}{63504}, \quad \frac{1381}{190512}.$$

They are positive. For $m = 6$ the convolution in (9.9) is positive, and for $m \geq 7$ both terms on the right are nonnegative with a positive convolution. Induction proves $c_m > 0$ for every m. Finally, $5f(z) \to \infty$ as $z \uparrow 1$, so $C(z) = 1 - 1/(5f(z)) \uparrow 1$; hence $\sum_{m\geq 1} c_m = 1$.

Moreover,

$$z\frac{d}{dz}\log(5f(z)) = \frac{5(z-C(z))}{1-z}.$$

Comparing coefficients gives

$$\ell\beta_\ell = 5\sum_{j>\ell} c_j, \qquad 0<\beta_\ell \le \frac{5}{\ell}. \tag{9.10}$$

Thus (8.2) is compound Poisson. Together with $(1-z)\sigma > 8/15$, equation (9.10) gives the all-order envelope as follows. The compound-Poisson cumulants satisfy

$$\kappa_s = \sum_{\ell\ge 1} \ell^s \beta_\ell z^\ell \le 5\sum_{\ell\ge 1} \ell^{s-1} z^\ell \le \frac{5(s-1)!}{(1-z)^s}.$$

After division by $s!\sigma^s$, this is

$$0 < \frac{\kappa_s}{s!\sigma^s} \le \frac{5}{s}\left(\frac{15}{8}\right)^s. \tag{9.11}$$

We also need two characteristic-function estimates. The integral representation

$$f(w) = \int_0^1 \frac{x^4}{1-wx}\,dx$$

implies that $|\phi_z(t)|$ decreases strictly on $[0,\pi]$. Indeed, after symmetrizing the double integral and putting $q = zx$, $r = zy$, and $y_0 = \cos t$, the derivative of the kernel with respect to y_0 has numerator

$$(1-q)(1-r)\mathcal{Q}(q,r) + 8qr(1-q)(1-r)(1-y_0) + 4qr(q+r)(1-y_0)^2,$$

where

$$\mathcal{Q}(q,r) = (q+r)\left(1-\frac{q+r}{2}\right)^2 + \frac{(q-r)^2(8-q-r)}{4} \ge 0.$$

The same rational saddle-band certificate proves, for the centered fourth moment μ_4,

$$\mu_4 < 18\sigma^4. \tag{9.12}$$

For an independent copy J', this gives

$$\mathbb{E}(J-J')^4 = 2\mu_4 + 6\sigma^4 < 42\sigma^4.$$

Using $1-\cos x \ge x^2/2 - x^4/24$, for $|t| \le 1/(2\sigma)$ we obtain

$$1-|\phi_z(t)|^2 \ge \sigma^2 t^2 - \frac{42\sigma^4 t^4}{24} \ge \frac{9}{16}\sigma^2 t^2. \tag{9.13}$$

Hence, in the variable $u = \sigma\sqrt{m}\,t$,

$$|\phi_z(t)|^m \le e^{-9u^2/32}. \tag{9.14}$$

At $t_0 = 1/(2\sigma)$, (9.13) gives

$$1-|\phi_z(t_0)|^2 \ge \frac{9}{64}.$$

Monotonicity therefore yields

$$|\phi_z(t)|^2 \le \frac{55}{64}, \qquad t_0 \le |t| \le \pi. \tag{9.15}$$

Now take the moving cutoff

$$U(m) = 6\left(\frac{m}{1000}\right)^{1/16}. \tag{9.16}$$

For a polynomial P, write

$$\langle P\rangle_G = \frac{1}{\sqrt{2\pi}}\int_{\mathbb{R}} e^{-u^2/2}|P(u)|\,du.$$

Let $\widehat{H}_s(u)$ be the positive formal majorants generated by

$$\exp\left(\sum_{j=1}^{8} A_j(u)\varepsilon^j\right) = \sum_{s\ge 0}\widehat{H}_s(u)\varepsilon^s.$$

They satisfy the same recurrence (9.7) with E_j replaced by A_j, and dominate the absolute coefficients of the exact $H_s(u,b)$.

The normalized error in (9.2) is divided into the following six explicit majorants:

$$E_{\text{core}}(m) = \sum_{\substack{12\le s\le 40\\ s\text{ even}}} m^{-s/2}\langle\widehat{H}_s\rangle_G, \tag{9.17}$$

$$E_{\text{point}}(m) \le \frac{4U(m)}{5}\left[\exp\left(\sum_{j=1}^{8} A_j(U(m))m^{-j/2}\right) - \sum_{s=0}^{40}\widehat{H}_s(U(m))m^{-s/2}\right], \tag{9.18}$$

and

$$E_{\ge 11}(m) \le \frac{20}{11(1-q_m)}\left(\frac{15}{8}\right)^{11} m^{-9/2}\mathbb{E}|Z|^{11}, \qquad q_m = \frac{15U(m)}{8\sqrt{m}}. \tag{9.19}$$

Here Z is standard normal. The retained polynomial outside the core is bounded by

$$E_{\text{poly}}(m) \le \sum_{j=0}^{5} m^{-j}\sum_{\ell}|h_{2j,\ell}|\frac{2}{\sqrt{2\pi}}\int_{U(m)}^{\infty} u^\ell e^{-u^2/2}\,du, \tag{9.20}$$

where $\widehat{H}_{2j}(u) = \sum_\ell h_{2j,\ell}u^\ell$. On $6 \le |u| \le 8$, the certified cumulant bounds sharpen (9.14) to $e^{-111u^2/250}$. Thus the remaining local Fourier tail is at most

$$E_{\text{near}}(m) \le \frac{2}{\sqrt{2\pi}}\left[\int_{U(m)}^{8} e^{-111u^2/250}\,du + \int_{\max(U(m),8)}^{\infty} e^{-9u^2/32}\,du\right], \tag{9.21}$$

with an empty first integral when $U(m) \ge 8$. Finally, (9.15) and $\sigma < 17$ give

$$E_{\text{far}}(m) \le \sigma\sqrt{2\pi m}\left(\frac{55}{64}\right)^{m/2}. \tag{9.22}$$

All exponentials and Gaussian tails in (9.17)–(9.22) are bounded by rational Taylor or Mills majorants in the certificate. At $m = 1000$, the contributions to a bound of the form C/m are respectively

component	contribution to C
E_{core}	0.0003482608957161
E_{point}	0.0005681456909691
$E_{\geq 11}$	0.0002847773304812
E_{poly}	0.0002761906358576
first integral in E_{near}	0.0000171697025750
second integral in E_{near}	0.0002701459962718
E_{far}	$< 2.1 \times 10^{-27}$

The displayed entries are decimal approximations to exact rational upper bounds; their exact sum is $0.001764690251870703\ldots < 0.001764690252$.

The moving cutoff makes these endpoint bounds uniform. For (9.17), multiplying by m leaves powers $m^{1-s/2}$ with $s \geq 12$. For the point tail, every omitted order $s \geq 41$ has degree at most $3s$, so its worst scaling exponent after multiplication by m is

$$\frac{17}{16} - \frac{5s}{16} < 0.$$

In (9.19), q_m is proportional to $m^{-7/16}$. For the polynomial Gaussian tail, the monomial $m^{-j}u^{\ell}$ has $\ell \leq 6j$. After multiplication by m, its worst logarithmic scaling exponent at the base cutoff $U = 6$ is

$$1 - j + \frac{\ell - 1}{16} - \frac{36}{16} \leq \frac{-21 - 10j}{16} < 0.$$

The Mills bounds for (9.21) have the form $mU^{-1}e^{-cU^2}$ and decrease from $U = 6$ or $U = 8$; the transition occurs at

$$m = \left\lceil 1000(4/3)^{16} \right\rceil .$$

The far bound decreases because its exponential factor dominates the square-root prefactor. These exact exponent checks prove

$$\left| \overline{p}_m(b) - \left(1 + \sum_{j=1}^{5} \frac{p_j(b)}{m^j} \right) \right| \leq \frac{C}{m}, \qquad C < 0.001764690252, \tag{9.23}$$

for every $m \geq 1000$ and each of the three displacements.

The retained probability polynomial differs from one by less than 0.009085849, so the passage to logarithms increases the absolute probability error by less than a factor two. The absolute weights in the logarithmic combination are $2 + 1 + 1 = 4$, and the smallest sample size is $k - 1$; the Fourier contribution is therefore at most $8C/(k - 1)$. The tails from the logarithm series, re-expansion of $(k \pm 1)^{-j}$, normalizing factors, and (6.4) all begin at order k^{-6} and are separately bounded by rational geometric majorants. At $k = 1001$ their total, together with the Fourier contribution, is

$$|R_k| < 1.419319083 \times 10^{-5}. \tag{9.24}$$

The right side of (8.11) is greater than

$$2.991014967 \times 10^{-5}. \tag{9.25}$$

After multiplication by k, every error term just listed is nonincreasing, whereas (8.11) is increasing. Therefore $\Delta(k, \mu) > 0$ throughout the compact band for every $k \geq 1001$.

10 Certified finite prefix and completion of the proof

It remains to cover the compact band for $2 \leq k \leq 1000$. Let

$$c(k, d) = 5^k B(k, d).$$

It satisfies the positive recurrence

$$(d + 5k)c(k, d) = (d - 1 + 5k)c(k, d - 1) + 5k\, c(k - 1, d), \qquad c(k, 0) = 1. \tag{10.1}$$

A 40-digit decimal computation with an explicit relative-error recurrence checks every point

$$2 \leq k \leq 1000, \qquad \frac{k}{2} < d < 12k. \tag{10.2}$$

There are $5,754,989$ such inequalities. The propagated relative distortion is less than 10^{-30}. The smallest computed relative margin is

$$0.0000322054177993889570397767399586094 72$$

at $(k, d) = (1000, 1868)$, which is larger than both the rounding budget and the acceptance threshold 10^{-25}. Thus every inequality in (10.2) is strict.

We can now prove Theorem 6.1. The region $k \geq 2d$ is covered by (7.13), and the region $d \geq 12k$ by (7.14). The remaining compact band is covered by the certified finite prefix when $k \leq 1000$ and by the effective Fourier estimate when $k \geq 1001$. The cases $k = 0, 1$, $d = 0$, and the right boundary are elementary; $k = 2$ is included by its independent exact proof. These regions contain every admissible pair (n, k) and give strictness whenever all three neighboring terms are positive. □

Combining Theorems 3.1, 4.1, and 6.1 with the known $r = 1, 2$ cases proves the positive direction of Theorem 1.1. Proposition 5.1 proves the converse.

11 Reproducibility and proof boundary

The proof has three kinds of evidence, which should not be conflated.

1. Sections 2–5 are ordinary symbolic proofs. The accompanying scripts independently regenerate the rational identities and coefficient signs.

2. Lemmas 7.1–7.3 and the coefficient inequalities (8.8)–(8.10) are finite symbolic certificates. Their programs reconstruct the relevant rational functions and verify all shifted monomial or Bernstein coefficients exactly.
3. Section 10 is an exhaustive finite computation with a proved rounding budget. It is used only for the explicit prefix (10.2). Section 9 proves the infinite remainder.

The complete internal verification map, interpreter versions, commands, expected terminal markers, and SHA-256 hashes are recorded in `verification/README.md`. The entry points are:

- `verification/verify_normalized_logconcavity.py`;
- `verification/verify_r4_weighted_cone.py`;
- `verification/verify_two_step_lc_preserver.py`;
- `verification/verify_small_excess_strips.py`;
- `verification/verify_sparse_cycle_k2.py`;
- `verification/verify_saddle_curvature.py`;
- `verification/verify_bulk_edgeworth_margin.py`;
- `verification/verify_bulk_second_order_margin.py`;
- `verification/verify_bulk_third_order_margin.py`;
- `verification/verify_bulk_fourth_order_margin.py`;
- `verification/verify_bulk_fifth_order_margin.py`;
- `verification/verify_renewal_reformulation.py`;
- `verification/verify_low_cumulant_envelope.py`;
- `verification/verify_characteristic_tail_constants.py`;
- `verification/verify_effective_fourier_remainder.py`;
- `verification/verify_bulk_finite_prefix.py`.

The reference environment is CPython 3.13.11 with SymPy 1.14.0. The aggregate PowerShell runner accepts an explicit Python interpreter and executes every program from the published `verification` directory. The fifth-order Bernstein certificate is the slowest symbolic step and required about 27 minutes in the recorded audit environment. Runtime is not part of the mathematical claim.

The machine checks establish the stated algebraic and finite inequalities, conditional on the correctness of the reductions and implementations. Sections 7 and 9 state the human reductions that connect those inequalities to Theorem 6.1; the programs reconstruct their rational certificates. These remain separate audit obligations. No claim of real-rootedness, total positivity, or Hankel-total positivity for the $r = 3, 4, 5$ rows is made here.

12 AI-assisted research and writing disclosure

Carptopus is the author and is responsible for the mathematical claims, source selection, verification materials, and publication decisions. OpenAI Codex was used to assist with symbolic exploration, exact verification programs, proof checking, literature searches, and drafting. All theorem statements and proof dependencies were reviewed against the archived source and verification materials before this draft was prepared.

References

[1] B. Deb and A. D. Sokal, *Higher-order Stirling cycle and subset triangles: Total positivity, continued fractions and real-rootedness*, arXiv:2507.18959v1 (2025). arXiv:2507.18959.

[2] P. J. Cameron, ed., *Problems from BCC30*, Problem 30.9, arXiv:2409.07216 (2024). arXiv:2409.07216.

[3] B. E. Sagan, *Inductive and injective proofs of log concavity results*, Discrete Mathematics **68** (1988), 281–292. doi:10.1016/0012-365X(88)90116-4.